AGK09

P1 Data Structs

n → number of data points
d → number of dimensions
k → number of centroids

data_points [n][d] > list of coordinates
centroids [n][d]
cluster_assigned [n] → cluster which point is assigned to

Core functionalities

- initialize the centroids() - k number of center points to act on the dataset
- calc_distance() - goes through the data set calculating the distance to all the centroids.
- update_centroid() - calculates the average pos of all point in the cluster. Moves center to avg.
- read csv()

Ü2 File → Super simplistic data set

The clustering is successful and it converges fast

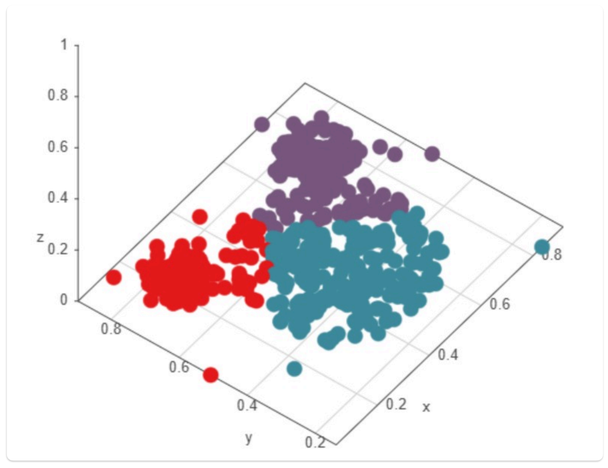
Nº of clusters	Nº of iterations
2	1
3	2

File → Mouse data set

The clustering is unsuccesful as it assumes the clusters are round like we discussed in class. So it misclassifies some of the head points in the 3 cluster one and it divides the face in the 4 cluster solution

3 clusters

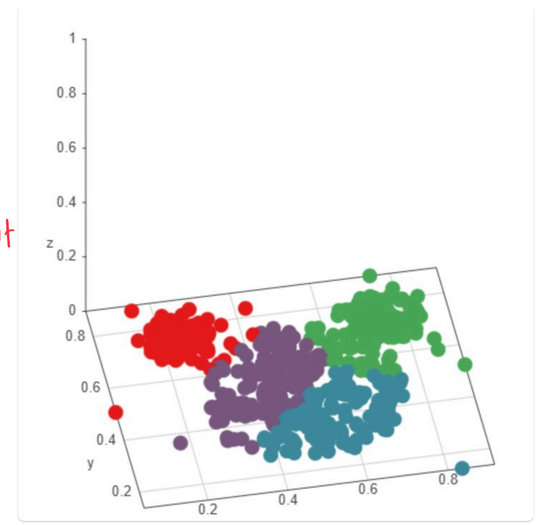
```
dinis@id6641:/mnt/c/Users/dinis/Desktop/kmeans-template$ ./kmeans 500 2 mouse.csv 3 euclidean
All tests passed successfully!
tests have passed
Converged in 6 iterations
Clustering result:
Centroid for cluster 0: 0.273141, 0.734209
Centroid for cluster 1: 0.713889, 0.694412
Centroid for cluster 2: 0.492566, 0.447994

Cluster composition (rows = predicted clusters, columns = true labels):

       T0   T1   T2
C0 |   25  100    0
C1 |   58    0  100
C2 |  215    0    0
```



4 clusters

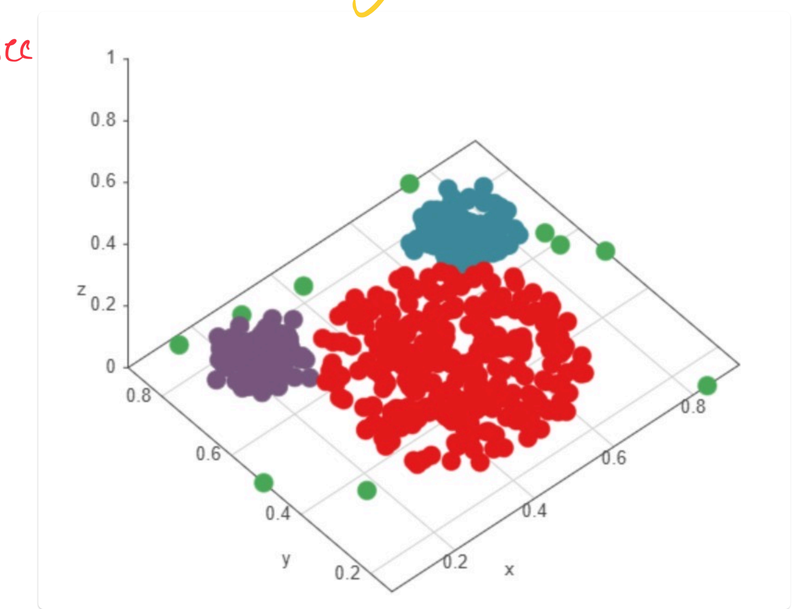
```
dinis@id6641:/mnt/c/Users/dinis/Desktop/kmeans-template$ ./kmeans 500 2 mouse.csv 4 euclidean
All tests passed successfully!
tests have passed
Converged in 16 iterations
Clustering result:
Centroid for cluster 0: 0.251767, 0.747288
Centroid for cluster 1: 0.423802, 0.542548
Centroid for cluster 2: 0.577183, 0.395627
Centroid for cluster 3: 0.727233, 0.708738

Cluster composition (rows = predicted clusters, columns = true labels):

       T0   T1   T2   T3
C0 |    6  100    0    4
C1 |  134    0    0    1
C2 |  115    0    0    1
C3 |   35    0  100    4
```

As we can see in this table T1 and T2 are correctly classified but T0 and T3 are not



Original

In this table we can see that clusters 1 e 2 are correct but 0 is very divided

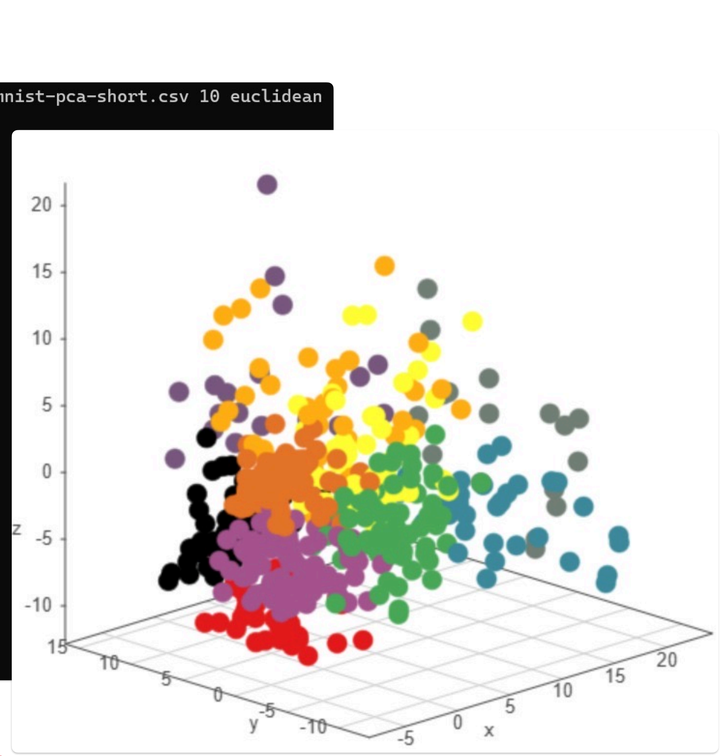
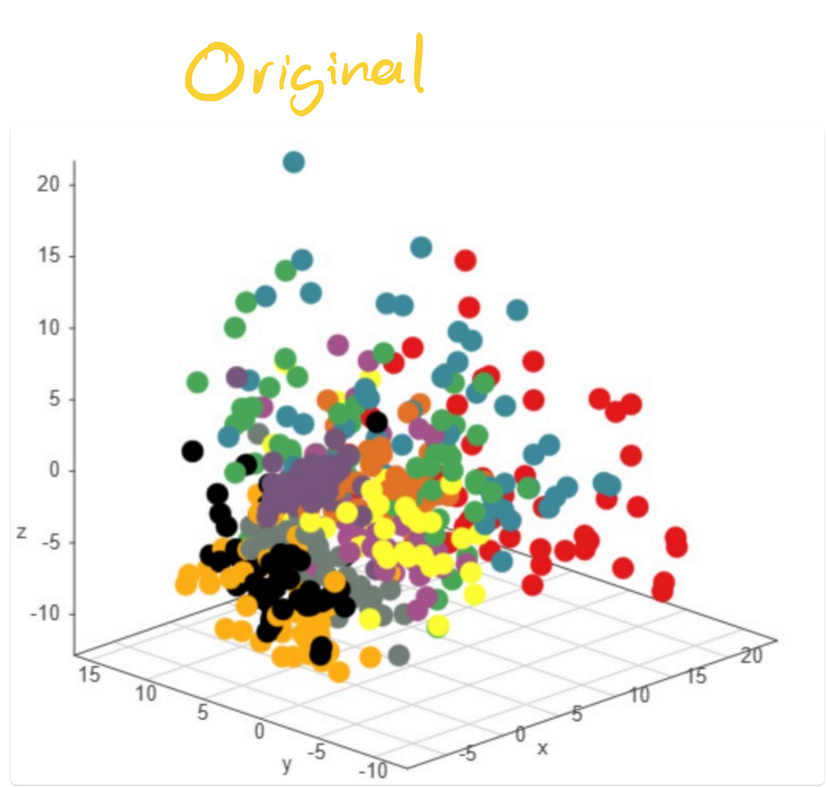


File → Mnist pca short dataset

The clustering is successful but is only a partial resemblense

```
dinis@id6641:/mnt/c/Users/dinis/Desktop/kmeans-template$ ./kmeans 500 4 mnist-pca-short.csv 10 euclidean
All tests passed successfully!
tests have passed
Converged in 12 iterations
Clustering result:
Centroid for cluster 0: -2.216942, 2.220972, -8.912678, 0.469269
Centroid for cluster 1: 4.967174, 9.935851, 4.140478, 0.264149
Centroid for cluster 2: 9.403892, -7.494795, -1.696619, 0.270813
Centroid for cluster 3: 0.023970, -6.665244, -0.504115, -5.209179
Centroid for cluster 4: 19.198066, 2.924842, -0.647078, -1.533515
Centroid for cluster 5: -4.735691, -1.700975, -3.403290, 1.665889
Centroid for cluster 6: -6.294868, -2.145458, 3.111138, 0.985154
Centroid for cluster 7: 2.394562, 1.929158, 5.596105, -6.493087
Centroid for cluster 8: 2.306875, -2.110466, 2.512110, 6.953731
Centroid for cluster 9: -2.844106, 5.824676, -2.716504, 0.481922

Cluster composition (rows = predicted clusters, columns = true labels):

       T0   T1   T2   T3   T4   T5   T6   T7   T8   T9
C0 |    0    0    0    0    6    0    0   22    0    9
C1 |    3    1    5    3    5    1    1    0    1    1
C2 |   20    0   13    0    0    0    0    0    0    0
C3 |    4    1    1   21    0   11    0    1   23    1
C4 |   17    0    0    0    1    0    0    0    0    0
C5 |    1    0    0    0   26   13    0   14    1   24
C6 |    0   64    3    0    3    5    6    3    8    0
C7 |    2    0    6   23    0    8    0    0    3    1
C8 |    2    0   21    0    1    1   38    0    0    0
C9 |    1    0    3    3   10    0    0   12    3   19
```

As we can see in the table, the clusters are all partially divided but with some resemblence to the original.



Original